DIPLOMA ELECTRONICS ENGINEERING • PROGRAM ELECTIVE

Consumer Electronics

A full handbook for course **6041C**. This is written as a screen-filling lesson file for revision, classroom teaching, service-technician understanding, and exam preparation. It covers audio devices, acoustic systems, video and TV systems, CCTV, and working principles of major home appliances.

SEMESTER	**6**
CREDITS	**4**
PERIODS	**60**
COS	**4**

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Official Syllabus Structure

The PDF defines this as a 60-period elective with one objective: awareness of commonly used consumer electronic equipment, physical quantities related to them, and working principles.

CO1 Audio Devices

14 hours. Sound fundamentals, microphones, headphones/headsets and loudspeakers.

CO2 Audio Systems

15 hours. Speaker baffles, indoor acoustics, PA system and digital audio players.

CO3 Video Systems

15 hours. Television fundamentals, TV technologies, cable/satellite TV and CCTV.

CO4 Home Appliances

14 hours. Induction cooker, microwave oven, washing machine, air conditioner and refrigerator.

Prerequisites

- ✓ Electronics and Electrical Engineering basics from Fundamentals of Electrical & Electronics Engineering.
- ✓ Electronic components and circuits from Electronic Circuits.
- ✓ Basic service awareness: voltage, current, resistance, power, earthing, insulation and safe isolation.

CO-PO Mapping

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						

CO3	2
CO4	2

MODULE 1 • CO1 • 14 HOURS

Audio Devices: Sound, Microphones, Headphones and Loudspeakers

Outcome: demonstrate the working principle of commonly used audio input and output devices.

1. Sound fundamentals

Sound is a longitudinal mechanical wave. Important quantities are amplitude, wavelength, frequency, time period, velocity, intensity and decibel level. Human hearing is roughly 20 Hz to 20 kHz, but practical audio systems are judged by usable frequency response, distortion and noise.

$$v = f\lambda, \quad T = 1/f, \quad dB = 10 \log(P_2/P_1), \quad SPL = 20 \log(p/p_0)$$

Frequency □□□□□□□□ pitch □□□□□; amplitude □□□□□□□□ loudness □□□□□. dB linear scale □□□□; logarithmic scale □□□.

2. Harmonics, octave and synthesis

A pure sine wave has one frequency. Musical notes and real voices contain fundamental plus harmonics. Octave means doubling or halving of frequency. Sound synthesis creates complex tones by adding waves with controlled amplitude and phase.

- Fundamental: lowest dominant frequency.
- Harmonic: integer multiple of the fundamental.
- Noise: random unwanted sound.

3. Microphones

A microphone is an acoustic-to-electrical transducer. Selection depends on sensitivity, frequency response, directionality, noise, ruggedness and application.

Type	Principle	Typical use
Moving coil / dynamic	Coil moves in magnetic field and induces voltage.	Stage, PA, rugged use.
Ribbon	Thin metal ribbon vibrates in magnetic field.	Studio, warm tone.

Crystal	Piezoelectric crystal generates voltage under stress.	Older/low-cost systems.
Capacitor / condenser	Capacitance changes with diaphragm movement; needs bias.	Studio, measurement.
Electret	Permanent charge material, compact condenser form.	Mobile, headset, tie-clip.
Wireless	Microphone + small RF transmitter + receiver.	Stage, teaching, events.

4. Headphones and headsets

Headphones convert electrical audio to sound close to the ear. A headset includes a headphone and microphone. Common types include moving iron, crystal, dynamic, electrostatic and electret. Controls may include volume, mute, track, call pickup and noise cancellation.

Service checks: broken cable near plug, open speaker coil, oxidised jack, faulty volume control, loose microphone boom and Bluetooth battery failure.

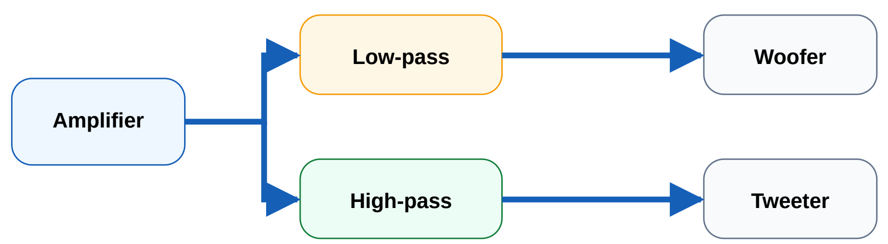
5. Loudspeaker principles

A loudspeaker is an electrical-to-acoustic transducer. The permanent magnet moving-coil speaker is common: current in the voice coil produces force in a magnetic field, moving the cone and air.

- ✓ Woofer: low frequency.
- ✓ Midrange: speech and main music range.
- ✓ Tweeter: high frequency.
- ✓ Crossover network: divides audio band between drivers.

6. Multi-way speaker system

One speaker cannot reproduce all frequencies efficiently. Multi-way systems combine woofer, midrange and tweeter with a crossover. A simple capacitor in series with a tweeter blocks low frequencies; inductors are used for low-pass sections.



Basic two-way crossover idea.

Exam answer frame: microphone comparison

Define microphone, state transducer action, draw simple principle diagram, compare moving coil/condenser/electret by principle, supply requirement, output level, ruggedness, cost and application. End with a selection example: dynamic for PA, condenser for studio, electret for mobile/headset.

MODULE 2 • CO2 • 15 HOURS

Audio Systems: Enclosures, Acoustics, PA Systems and Digital Audio

Outcome: understand the characteristics and features of audio systems used in rooms, auditoriums, studios and public address installations.

Speaker baffles and enclosures

A naked speaker cone produces sound from front and rear. The rear wave is out of phase with the front wave and can cancel bass. A baffle/enclosure reduces cancellation and controls low-frequency response.

System	Working idea	Notes
Open baffle	Large board separates front and rear waves.	Simple, poor compact bass.
Sealed enclosure	Air inside acts as spring.	Tight bass, lower efficiency.
Bass reflex	Port uses rear energy to reinforce bass near tuning frequency.	Common in home speakers.
Folded horn	Horn path is folded inside cabinet to improve acoustic coupling.	High efficiency, large size.

Indoor acoustics

Room sound depends on reflection, absorption, diffusion and reverberation. Hard walls reflect sound; curtains, carpets and acoustic panels absorb sound.

- ✓ Living room: moderate absorption, avoid echo and booming bass.
- ✓ Auditorium: controlled reverberation and uniform coverage.
- ✓ Studio: low noise, controlled reflections, accurate monitoring.
- ✓ Too much reflection reduces speech clarity.

Reverberation time = time for sound level to decay by 60 dB after source stops.

Public Address System

A PA system distributes speech or music to a large audience. It must provide adequate loudness, intelligibility, reliability and safe wiring.

1. Input transducer: wired/wireless microphone.
2. Mixer or preamplifier: level control and tone adjustment.
3. Power amplifier: drives speaker lines.
4. Loudspeaker network: column speakers, horn speakers or ceiling speakers.
5. Protection: proper earthing, overload protection, speaker line matching.

Megaphone

A megaphone is a portable PA system with microphone, amplifier, battery and horn speaker in one unit. The horn improves directivity and acoustic efficiency.

Faults: weak battery, loose switch, cracked horn, faulty microphone capsule, distorted amplifier stage.

Digital audio players

Digital audio stores sound as samples. In playback, memory or optical disc data is read, decoded, converted to analog using DAC, filtered, amplified and sent to headphones or speakers. MP3 reduces file size using perceptual compression.

Optical recording on disc: data is represented by pits and lands. A laser pickup reads reflected light changes. Servo systems maintain focus and tracking.

MP3: removes less audible information, so it is lossy. Higher bit rate usually means better quality and larger file size.

Digital audio playback chain.

MODULE 3 • CO3 • 15 HOURS

Video Systems: TV Fundamentals, TV Types, Cable/Satellite TV and CCTV

Outcome: explain the concepts used in video display and video distribution systems.

Television fundamentals

- ✓ **Persistence of vision:** eye retains an image for a short time, allowing motion illusion.
- ✓ **Aspect ratio:** width-to-height ratio such as 4:3 or 16:9.
- ✓ **Resolution:** number of pixels in image; higher resolution gives more detail.
- ✓ **Pixel:** smallest picture element.
- ✓ **Hue:** colour type; **brightness:** lightness; **saturation:** colour intensity.
- ✓ **Luminance:** brightness information; **chrominance:** colour information.

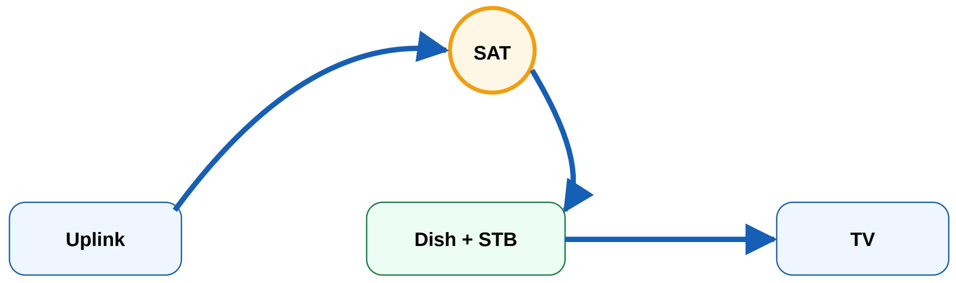
TV technology comparison

Type	Working idea	Benefits / limits
LCD TV	Liquid crystals control light from backlight.	Thin, efficient; black level depends on backlight.
LED TV	LCD panel with LED backlighting.	Better efficiency and slim design.
Plasma TV	Gas cells emit light when excited.	Good motion and contrast; high power, older technology.
Projection TV	Image engine projects picture onto screen.	Large image; needs alignment and lamp/engine care.

3D TV	Separate left/right eye images create depth perception.	Needs glasses or special display system.

Cable and satellite TV

A satellite TV system uses uplink station, satellite transponder, downlink dish, LNB, set-top box and television. DTH means direct-to-home reception. Cable TV uses a headend/distribution station and coaxial/fibre network to supply many subscribers.



Basic DTH satellite TV chain.

Internet through cable TV network

Modern cable networks can carry television and data using frequency division. Cable modem converts RF data into Ethernet/Wi-Fi data. The headend connects to internet backbone and manages subscriber access.

Important service points: splitter loss, bad connectors, low signal level, water ingress in coaxial cable and poor grounding.

CCTV systems

CCTV is a closed video surveillance system. A basic system has camera, transmission medium, DVR/NVR, monitor, storage and power supply. IP CCTV uses network communication; analog CCTV uses coaxial video signals.

Camera	Use	Key point
Bullet	Outdoor wall/ceiling mounting.	Visible deterrent, directional.
Dome	Indoor/outdoor ceiling use.	Compact and tamper-resistant.
Covert	Hidden surveillance.	Small camera concealed in object.
C-Mount	Lens-change applications.	Flexible focal length.
Night vision	Low-light monitoring.	Uses IR LEDs or low-lux sensor.

Day/Night	Changing light conditions.	Switches between colour and monochrome.

CCTV answer □□□□□□□□□□□ block diagram + camera type + storage + power supply + application + one maintenance point □□□□□□□□□ answer complete □□□ □□□□□□□□.

MODULE 4 • CO4 • 14 HOURS

Home Appliances: Induction Cooker, Microwave Oven, Washing Machine, AC and Refrigerator

Outcome: explain the working principle of frequently used domestic appliances.

Induction cooker

An induction cooker heats the vessel directly by electromagnetic induction. AC mains is rectified, filtered and converted to high-frequency AC by an inverter. The work coil produces alternating magnetic field, inducing eddy currents in a ferromagnetic vessel. The vessel heats due to I^2R loss.

1. AC input and EMI filter.
2. Rectifier and DC link capacitor.
3. High-frequency inverter using power device.
4. Work coil and suitable vessel.
5. Temperature/current sensing and control board.

Advantages: fast heating, efficient, clean surface, quick control. **Disadvantages:** needs compatible vessel, electronic repair complexity, EMI and fan noise.

Microwave oven

A microwave oven uses electromagnetic waves around 2.45 GHz to agitate polar water molecules in food. The magnetron generates microwave energy, waveguide carries it to cavity, and turntable/stirrer improves uniform heating.

Safety: never operate with door interlock bypassed, damaged cavity, exposed HV section or removed shielding. High-voltage capacitor can retain dangerous charge after unplugging.

Common blocks: mains filter, transformer/inverter, rectifier, magnetron, waveguide, cavity, cooling fan, turntable motor, lamp, control board and door interlocks.

Washing machine

A washing machine performs filling, washing/agitation, draining, rinsing and spinning. Main parts include inlet valve, water-level sensor, drum, motor, belt/direct drive, drain pump, heater in some models and electronic controller.

Type	Feature
Semi-automatic	Separate wash and spin tubs; manual transfer.
Fully automatic top-load	Vertical drum, automatic fill/wash/rinse/spin.
Fully automatic front-load	Horizontal drum, efficient washing, lower water use.

Air conditioner

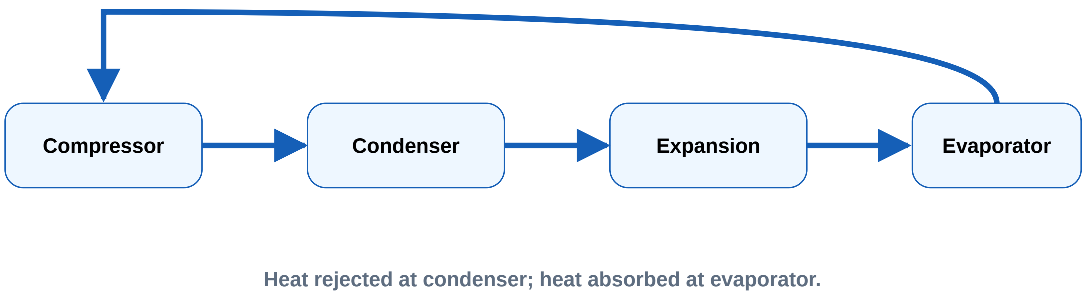
Air conditioning controls temperature, humidity, air circulation and cleanliness. Major parts: compressor, condenser, expansion device, evaporator, blower, filter, fan motors and control electronics.

Ton of refrigeration: cooling capacity equal to the heat removal rate of melting one ton of ice in 24 hours. Practically, 1 TR ≈ 3.517 kW ≈ 12000 BTU/hr.

- ✓ Unitary/window AC: compact single unit.
- ✓ Split AC: indoor and outdoor units connected by refrigerant pipes.
- ✓ Centralized AC: large plant for building-level cooling.

Refrigerator

A domestic refrigerator uses vapour-compression refrigeration. The compressor raises refrigerant pressure and temperature; condenser rejects heat; expansion device lowers pressure; evaporator absorbs heat from cabinet.



Vapour compression refrigeration cycle.

Service-minded appliance troubleshooting

- ✓ Start with symptom, supply, fuse, switch and connector checks.

- ✓ Look for overheating, loose terminals, moisture and damaged insulation.
- ✓ Verify sensors before condemning control PCB.
- ✓ Never bypass interlocks or protective devices for normal use.
- ✓ Record model, voltage rating, power rating and fault condition.

Formula Bank and Fast Revision Tables

Use this part for last-day exam revision and numerical support.

Sound and audio

$v = f\lambda$

$T = 1/f$

Power ratio in dB = $10 \log_{10}(P_2/P_1)$

Voltage ratio in dB = $20 \log_{10}(V_2/V_1)$

One octave increase = frequency × 2

Appliance and cooling

Electrical power $P = VI \cos\varphi$ (AC load)

Heat energy $Q = mc\Delta T$

1 ton of refrigeration ≈ 3.517 kW ≈ 12000 BTU/hr

Induction heating loss in vessel ≈ I^2R

Common comparisons

Pair	Difference
Headphone vs headset	Headphone only output; headset includes microphone also.
Luminance vs chrominance	Luminance carries brightness; chrominance carries colour information.
LCD vs LED TV	LED TV is LCD technology using LED backlight; older LCD may use CCFL backlight.
AC vs refrigerator	Both use refrigeration cycle; AC cools room air, refrigerator cools insulated cabinet.
Microwave vs induction	Microwave heats food by dielectric heating; induction heats vessel by eddy currents.

Practice Question Bank

Questions are arranged module-wise. Answer in diagram + principle + working + application format wherever possible.

Module 1

1 Define wavelength, frequency, time period and velocity of sound.

2 Explain decibel level in acoustic measurement.

3 Compare moving coil and condenser microphones.

4 Explain the working of permanent magnet loudspeaker.

5 What is crossover network in multi-way speaker system?

Module 2

6 Explain the function of speaker baffle.

7 Compare bass reflex and folded horn systems.

8 Explain reflection, absorption and reverberation.

9 Draw and explain PA system block diagram.

10 Explain digital audio playback from optical disc.

Module 3

11 Define aspect ratio, resolution, pixel, hue and saturation.

12 Compare LCD, LED and Plasma TV.

13 Explain DTH system with block diagram.

14 Explain CCTV system with block diagram.

15 List different CCTV camera types and applications.

Module 4

16 Explain induction cooker with block diagram.

17 Explain microwave oven working principle and safety instructions.

18 Explain washing machine block diagram and sequence.

19 Define ton of refrigeration and list AC types.

20 Explain domestic refrigerator using block diagram.

Model Question Paper

Practice paper prepared from the official syllabus structure. Use it for timed revision.

Course: 6041C Consumer Electronics
Time: 3 Hours Maximum Marks: 100

PART A - Short Answer Questions
1. Define frequency and wavelength of sound.
2. What is an octave in music?
3. List any four types of microphones.
4. What is the purpose of a crossover network?
5. Define reverberation.
6. List the main blocks of a PA system.
7. What is MP3 format?
8. Define pixel and aspect ratio.
9. What is DTH?
10. Name four CCTV camera types.

PART B - Short Essay Questions

11. Explain the working principle of moving coil microphone.
12. Compare headphone and headset with examples.
13. Explain speaker baffles and bass reflex enclosure.
14. Draw and explain a public address system.
15. Explain television fundamentals: hue, brightness, saturation, luminance and chrominance.
16. Compare LCD TV, LED TV and Plasma TV.
17. Explain cable and satellite TV distribution.
18. Explain the working of induction cooker.
19. Explain microwave oven with safety instructions.
20. Explain refrigerator block diagram and working.

PART C - Essay Questions

21. Explain different microphone types and their applications.
22. Explain indoor acoustics for living room, auditorium and studio.
23. Explain CCTV system with block diagram and camera classification.
24. Explain air conditioner and refrigerator working principles with cooling cycle.

Answer Key and Marking Guide

Use these points to build standard exam answers. Do not write one-line answers for block diagram questions.

Part A key points

1. Frequency = cycles per second; wavelength = distance between successive compressions/rarefactions.
2. Octave means doubling/halving of frequency.
3. Moving coil, ribbon, crystal, condenser, electret, wireless.
4. Crossover divides audio frequencies among woofer, midrange and tweeter.
5. Reverberation is persistence of sound due to repeated reflections.
6. Microphone, mixer, amplifier, speaker, power supply.
7. MP3 is lossy compressed digital audio format.
8. Pixel = picture element; aspect ratio = width:height.
9. DTH = direct-to-home satellite TV reception.
10. Bullet, dome, covert, C-mount, night vision, day/night.

How to answer big questions

Induction cooker: draw AC input → rectifier → inverter → work coil → vessel. Explain eddy current heating, sensor feedback, advantages and disadvantages.

Microwave oven: draw power supply → magnetron → waveguide → cavity. Explain dielectric heating, turntable, interlocks and safety.

CCTV: draw camera → cable/network → DVR/NVR → monitor/storage. Explain working, camera types and applications.

PA system: draw microphone → preamplifier/mixer → power amplifier → speakers. Explain function of each block and feedback prevention.

Text / References and Online Resources

Books

- Bali S P, *Consumer Electronics*, Pearson Education India.
- Gulati R R, *Modern Television Practices*, New Age International.
- Gupta R G, *Audio Video Systems*, Tata McGraw Hill.
- Amit Dhir, *The Digital Consumer Technology Handbook*, Xilinx.
- Gupta R G, *Television Engineering and Video Systems*, Tata McGraw Hill.

Official online topics

- Induction cooktops and induction heating.
- TV port guide and cable TV connection basics.
- LCD, LED and Plasma TV differences.
- Plasma TV explanation and display basics.

References syllabus-il □□□□□□□□ books/resources □□□□□□□□□□□□□. Exam answer □□□□□□□ textbook terminology use □□□□□□□□.